SS-C-192g
December 4, 1964

SUPERSEDING
Int. Fed. Spec. SS-C-00192f (COM-NBS)
June 12, 1963 and
Fed. Spec. SS-C-192d
June 27, 1960

FEDERAL SPECIFICATION

# CEMENT, PORTLAND

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.** This specification covers eight types of Portland cement—five types without air-entraining addition, each designated by number, and three corresponding types with air-entraining addition, each designated by inclusion of the letter "A" in the type number.

**1.2 Classification.**

**1.2.1** *Types.* Portland cement shall be of the following types, as specified (6.2):

Type I—For use in general concrete construction when the special properties specified for types II, III, IV, and V are not required.

Type IA—Air entraining cement for the same uses as type I.

Type II—For use in general concrete construction and in construction exposed to moderate sulfate action.

Type IIA—Air-entraining cement for the same uses as type II.

Type III—For use when high early strength is required.

Type IIIA—Air-entraining cement for the same use as type III.

Type IV—For use when low heat of hydration is required (6.3).

Type V—For use when high sulfate resistance is required (6.3).

## 2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

*Federal Specification*:

UU-S-48—Sacks; Paper, Shipping.

*Federal Standards*:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 158—Cements, Hydraulic; Sampling, Inspection and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)



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*Military Standards:*

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Materials.** Portland cement may be made from any selection of raw materials and by any process that will yield a product conforming to the requirements for the type specified.

**3.2 Chemical and physical requirements.** Portland cement shall conform to the applicable requirements in tables I and II.

TABLE I.—*Chemical requirements*[1]

| | Types I and IA | Types II and IIA | Types III and IIIA | Type IV | Type V |
|---|---|---|---|---|---|
| | *Percent* | *Percent* | *Percent* | *Percent* | *Percent* |
| Silicon dioxide ($SiO_2$), minimum | — | 21.0 | — | — | — |
| Aluminum oxide ($Al_2O_3$), maximum | 7.5 | 6.0 | 7.5 | 6.0 | — |
| Ferric oxide ($Fe_2O_3$), maximum | 6.0 | 6.0 | 6.0 | 6.5 | — |
| Magnesium oxide (MgO), maximum | 5.0 | 5.0 | 5.0 | 5.0 | 4.0 |
| Sulfur trioxide ($SO_3$) | | | | | |
| When $3CaO \cdot Al_2O_3$ is 8.0 percent or less, maximum | 2.5 | 2.5 | 3.0 | 2.3 | 2.3 |
| When $3CaO \cdot Al_2O_3$ is more than 8.0 percent, maximum | 3.0 | — | 4.0 | — | — |
| Loss on ignition, maximum | 3.0 | 3.0 | 3.0 | 2.5 | 3.0 |
| Insoluble residue, maximum | 0.75 | 0.75 | 0.75 | 0.75 | 0.75 |
| Tricalcium silicate[2] ($3CaO \cdot SiO_2$), maximum | — | — | — | 35 | — |
| Dicalcium silicate[2] ($2CaO \cdot SiO_2$), minimum | — | — | — | 40 | — |
| Tricalcium aluminate[2] ($3CaO \cdot Al_2O_3$), maximum | 15.0 | 8.0 | 15.0[3] | 7.0 | 5.0 |
| Tricalcium aluminate plus tricalcium silicate[2] ($3CaO \cdot Al_2O_3 + 3CaO \cdot SiO_2$), maximum | — | 58[4] | — | — | 55[4] |
| Tetracalcium aluminoferrite plus twice the tricalcium aluminate[2] ($4CaO \cdot Al_2O_3 \cdot Fe_2O_3$)+2($3CaO \cdot Al_2O_3$) or solid solution ($4CaO \cdot Al_2O_3 \cdot Fe_2O_3 + 2CaO \cdot FE_2O_3$), as applicable, maximum | — | — | — | — | 20.0 |
| Alkalies (summation percentage $Na_2O$+ 0.658 percent $K_2O$)[5], maximum | 0.60 | 0.60 | 0.60 | 0.60 | 0.60 |

[1] Referee methods shall be used in case of dispute, or when the results obtained by optional methods indicate a result so close to the specification limit that meeting or failing to meet the specification requirements is in doubt. When referee methods are required, all determinations of constituents involved in the computations, shall be made entirely by referee methods, and the separations prior to these determinations shall be made by referee methods.

Examples. If a cement which is supposed to meet type I requirements contains 2.59 percent $SO_3$ and 7.0 percent tricalcium aluminate, the cement shall not be rejected unless values for $SO_3$ and $Al_2O_3$ have been verified by referee methods, and all prior separations made by referee methods. Likewise, when meeting the tricalcium silicate requirement for type IV is in doubt, it shall not be rejected unless CaO, $SiO_2$, $Al_2O_3$ and $SO_3$ have been determined by referee methods.

[2] The expressing of chemical limitations by means of calculated assumed compounds does not necessarily mean that the oxides are actually or entirely present as such compounds.

When the ratio of percentages of aluminum oxide to ferric oxide is 0.64 or more, the percentages of tri-



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calcium silicate, dicalcium silicate, tricalcium aluminate and tetracalcium aluminoferrite shall be calculated from the chemical analysis as follows:

Tricalcium silicate = (4.071 × percent CaO) − (7.600 × percent $SiO_2$) − (6.718 × percent $Al_2O_3$) − (1.430 × percent $Fe_2O_3$) − (2,852 × percent $SO_3$)

Dicalcium silicate = (2.867 × percent $SiO_2$) − (0.7544 × percent $C_3S$)

Tricalcium aluminate = (2.650 × percent $Al_2O_3$) − (1692 × percent $Fe_2O_3$)

Tetracalcium aluminoferrite = 3.043 × percent $Fe_2O_3$

When the alumina-ferric oxide ratio is less than 0.64, a calcium aluminoferrite solid solution (expressed as ss($C_4AF+C_2F$) is formed. Contents of this solid solution and of tricalcium silicate shall be calculated by the following formulas:

ss($C_4AF+C_2F$) = (2.100 × percent $Al_2O_3$) + (1.702 × percent $Fe_2O_3$)

Tricalcium silicate = (4.071 × percent CaO) − (7.600 × percent $SiO_2$) − 4.470 × percent $Al_2O_3$) − (2.859 × percent $Fe_2O_2$) − (2.852 × percent $SO_3$).

No tricalcium aluminate will be present in cements of this composition. Dicalcium silicate shall be calculated as previously shown.

In the calculation of $C_3A$, the values of $Al_2O_3$ and $Fe_2O_3$ determined to the nearest 0.01 percent shall be used. In the calculation of other compounds the oxides determined to the nearest 0.1 percent shall be used.

Values for $C_3A$ and for the sum of $C_4AF + 2C_3A$ shall be reported to the nearest 0.1 percent. Values for other compounds shall be reported to the nearest 1 percent.

[3] When moderate sulfate resistance is required for type III cement, tricalcium aluminate may be limited to 8 percent. When high sulfate resistance is required, the tricalcium aluminate may be limited to 5 percent.

[4] This requirement applies only when specifically requested in the invitation for bids.

[5] This limit applies only when "low alkali" cement is specified.

TABLE II.—*Physical requirements*

| Types | I | IA | II | IIA | III | IIIA | IV | V |
|---|---|---|---|---|---|---|---|---|
| Fineness, air permeability: | | | | | | | | |
| Average value, minimum, sq.cm./g. | 2,800 | 2,800 | 2,800 | 2,800 | — | — | 2,800 | 2,800 |
| Minimum value, any one sample in average, sq.cm./g. | 2,600 | 2,600 | 2,600 | 2,600 | — | — | 2,600 | 2,600 |
| Soundness, autoclave expansion, maximum, percent | 0.80 | 0.80 | 0.80 | 0.80 | 0.80 | 0.80 | 0.80 | 0.80 |
| Time of setting, Gillmore test: | | | | | | | | |
| Initial, minimum, minutes | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 |
| Final, maximum, hours | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 |
| Compressive strength:[1] (min. p.s.i.) | | | | | | | | |
| 1 day | — | — | — | — | 1,700 | 1,300 | — | — |
| 3 days | 1,200 | 900 | 1,000 | 750 | 3,000 | 2,500 | — | — |
| 7 days | 2,100 | 1,500 | 1,800 | 1,400 | — | — | 800 | 1,500 |
| 28 days | 3,500 | 2,800 | 3,500 | 2,800 | — | — | 2,000 | 3,000 |
| Air entrainment: | | | | | | | | |
| No addition used, maximum percent by volume | 12.0 | — | 12.0 | — | 12.0 | — | 12.0 | 12.0 |
| Using air-entraining addition, percent by volume | — | 19.0 ± 3.0 | — | 19.0 ± 3.0 | — | 19.0 ± 3.0 | — | — |
| Heat of hydration:[2] | | | | | | | | |
| 7 days, maximum, cal./g. | — | — | 70 | 70 | — | — | 60 | — |
| 28 days, maximum, cal./g. | — | — | 80 | 80 | — | — | 70 | — |
| False set[3], early stiffening of cement paste, ratio of final to initial penetration, percent, minimum | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |

[1] The strength at any age shall be greater than the strength at the next preceding age. Unless otherwise



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specified, strength tests for types I, IA, II, and IIA, will be made only at 3 days and 7 days. If at the option of the purchaser, a 7 day test is required in types III and IIIA cement, the strength at 7 days shall be greater than at 3 days.

[2] These requirements apply only when specifically requested; when the heat of hydration requirements are specified for types II and IIA, the strength requirements shall be 80% of the values listed in table II.

[3] This requirement applies only when specified (6.2).

3.3 **Processing additions.** Only the following materials shall be added during grinding of the clinker: (1) Water and/or untreated calcium sulfate in amounts such that the limits shown in table I for loss on ignition and sulfur trioxide are not exceeded, and (2) At the option of the manufacturer, nondeleterious materials other than water and/or calcium sulfate in amounts less than one percent. When specified (6.2), the manufacturer shall furnish a written statement certifying that less than one percent of nondeleterious material other than water and calcium sulfate or calcium sulfate, and no deleterious material has been added.

3.4 **Air-entraining addition.** For air-entraining cement there shall be interground with the clinker an addition of such a nature and in such an amount that the resulting product complies with the applicable requirements of tables I and II. When specified (6.2), the manufacturer shall furnish a written statement giving the source, amount, and brand name of the addition used. When specified (6.2), the manufacturer shall also furnish test data showing that an air-entraining portland cement, (1) produced with this air-entraining addition, (2) meeting the applicable requirements of tables I and II, and (3) manufactured not more than three years prior to the date of request, had a durability factor of not less than 80 when tested in accordance with the procedures of method 4201 of Fed. Test Method Std. No. 158.

3.5 **Identification marking.** The brand, type, and net weight shall be clearly marked on each bag.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection responsibility.** Cement for use on Federal Government projects will be sampled at the mill and will be tested for conformance to the specified requirements by the National Bureau of Standards, or other designated Government agency. Sampling and testing by a Government agency may be waived at the discretion of the purchasing agency, and the cement accepted upon the manufacturer's certification of compliance with specification requirements accompanied by a mill test report.

4.2 **Sampling and storage.** Sampling and storage shall be in accordance with section 4 of Fed. Test Method Std. No. 158.

4.2.1 *Lot.* Unless otherwise specified (6.2), a lot shall consist of each type of cement submitted for inspection at the same time.

4.2.2 *Sampling for examination of packing.* Packed cement shall be sampled in accordance with level S-3 in accordance with MIL-STD-105.

4.2.3 *Sampling for test.* Portland cement shall be sampled in accordance with section 4 of Fed. Test Method Std. No. 158.

4.3 **Inspection.** Packed cement selected in 4.2.2 shall be examined in accordance with table III. The acceptable quality levels (AQL's) for defects listed in table III shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects in accordance with MIL-STD-105.

TABLE III—*Classification of defects*

| Defects | Major | Minor |
|---|---|---|
| Bags not of the grade specified | X | |
| Bags not fully closed ..... | X | |
| Bags split or torn ..... | X | |
| Hole in bags ..... | X | |
| Marking omitted or illegible .. | X | |
| Marking incomplete ..... | | X |
| Marking in wrong location .... | | X |



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4.4 **Tests.** Chemical tests for the requirements in table I and the tests for physical requirements in table II shall be conducted in accordance with the applicable test methods of Fed. Test Method Std. No. 158.

4.5 **Rejection and retests.** The cement shall be rejected if it fails to meet any of the requirements of this specification. Cement remaining in storage prior to shipment for a period longer than 6 months after test may be retested and shall be rejected if it fails to meet any of the requirements of this specification. Cement having an air content between 12 and 16 percent as determined by the air-entrainment test may be accepted at the discretion of the purchaser when air-entraining concrete is to be used and the air content of the concrete is controlled at the mixer. Cement failing to conform to the limit on autoclave expansion test may be accepted if it passes in a retest of new samples taken within 28 days of the time of the original sampling. The provisional acceptance of the cement at the mill shall not deprive the purchaser of the right of rejection on a retest of autoclave expansion at the time of the delivery of the cement to the purchaser.

4.6 **Inspection of preparation for delivery.** The packing and marking of the cement shall be examined to determine compliance with the requirements of section 5. The brand, type, and net weight shall be clearly marked on the bags.

## 5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the levels of packaging shall be in accordance with Fed. Std. No. 102.)

5.1 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Cement in quantities of 94 pounds net shall be packed in bags conforming to UU-S-48, X grade.

5.1.2 *Level B.* Cement in quantities of 94 pounds net shall be packed in bags conforming to UU-S-48, L/W grade.

5.1.3 *Level C.* Cement shall be packed to insure carrier acceptance and safe delivery to destinations in containers complying with the rules and regulations applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to any special marking specified in the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to any special markings specified in the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

## 6. NOTES

6.1 **Intended use.** Portland cement is intended for use in concrete construction. With the possible exception of type IV (which has a low rate of increase of strength), all of these cements are suitable for use in general concrete construction. There are specific conditions which may make the use of type II, III, or V more desirable than type I.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type of cement required (1.2), when not specified, type I should be furnished.

(*c*) When required, a written statement on nondeleterious material (3.3).

(*d*) When required, a written statement on the source, amount, and brand name of air-entraining addition (3.4).

(*e*) When required, test data on air-entraining addition (3.4).



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(*f*) Size of lot, if different from 4.2.1.

(*g*) Level of packing required (5.1).

(*h*) When required, reduced maximum limit on $C_3A$ for type III (table I, note 3).

(*i*) When required, limit on the sum of tricalcium silicate plus tricalcium aluminate, types II and V (table I, note 4).

(*j*) When required, limit on alkalies (table I, note 6).

(*k*) When required, limit on heat of hydration (table II, note 2).

(*l*) When required, limit on false set (table II, note 3).

(*m*) When required, compressive strength tests for any age other than those shown in table II (note 1).

6.3 Cements conforming to the requirements for types IV and V usually are not carried in stock. In advance of specifying their use, the purchaser or his representative should ascertain whether these types of cement are, or can be made available.

6.4 **Transportation description.** The transportation descriptions and minimum weights applicable to this commodity are:

*Rail*:

Cement, Portland
Carload minimum weight 50,000 pounds.

*Motor*:

Cement, Portland
Truckload minimum weight 50,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.5 This material is similar to the cement described in ASTM C150 and C175.

MILITARY INTEREST:

Army—MO

Navy—Docks

Air Force—69

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Copies of this specification may be purchased for ⁻ cents each.



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